BUILD A CUBESAT

VERSION 0.2 · 2026-10-29

Rocinante / Roci – Software Stack Naming Note

flight software naming convention

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This note documents the naming convention for **Rocinante**, the product identity of BAC's flight software stack. It applies the naming, versioning and glyph conventions of the BAC Project & Tooling Guide (§2.2, §2.3, §2.6) and the voice of the BAC Identity Guide (§2); the Rocinante marks are defined in the Identity Guide §8 and covered by the BAC Trademark Policy §2. The guides live in bac-design (https://github.com/buildacubesat/bac-design).

1 What Rocinante Is

Rocinante is the name of the software stack running on BAC's onboard computers. It spans the three node classes of the BAC Project Overview §5.2:

- **Small nodes** – Zephyr RTOS on small, energy-efficient MCUs for deterministic subsystem control: always-on nodes such as the EPS MCU, and sensor heads.
- **Medium nodes** – Zephyr and NASA F Prime on larger MCUs for more demanding flight tasks such as FDIR and COMMS.
- **Large nodes** – Linux and NASA F Prime on higher-capability compute hardware such as the Raspberry Pi CM5; payload nodes.

The MicroMod carrier serves small and medium nodes, the CM5 carrier large nodes. STM32F405 processor boards are supported first, STM32U5 later. Nodes talk through a shared protocol boundary – CSP over CAN – rather than running one framework everywhere. The ground segment is out of scope for this note.

Rocinante is a **product identity**, not a repository name. It names the stack as a coherent thing a contributor or user can refer to – "the Rocinante stack handles X" – independent of where the code lives.

2 Naming Rule

CONTEXT	FORM
First mention in a document or conversation	**Rocinante**
Subsequent mentions in the same document or conversation	**Roci**
C and C++ identifiers, files, modules	lowercase roci_ prefix
Kconfig symbols	CONFIG_ROCI_*
F Prime namespace	Roci

This mirrors how **Build a CubeSat** shortens to **BAC** after first reference, and how bac is used as a lowercase code and file name prefix (Project & Tooling Guide §2.3). Where a language or framework fixes the casing – uppercase Kconfig symbols, PascalCase F Prime namespaces – the prefix follows it. Within one identifier the casing is never mixed, and prose never writes Roci_, rOci or ROCI.

Examples:

- Prose, first mention: *"The EPS node runs Rocinante on a small node."*
- Prose, thereafter: *"Roci handles telecommanding over CSP."*
- Code: roci_eps, roci_hk_collector, CONFIG_ROCI_HEARTBEAT, Roci::Topology.
- Commit and PR titles: treat as a proper noun, capitalized normally in a sentence – *"Add Roci telemetry rate group for BNO086."*

3 Relationship to bac-software

Rocinante is not a repository rename. The question is settled by the bac-software docs split (bac-software #105):

- **Repository:** bac-software keeps its name (Project & Tooling Guide §2.8).
- **Product identity inside the repository:** Rocinante / Roci, in namespaces, documentation headers and prose.
- **Document IDs** stay BAC-FSW-* and BAC-SBC-*, with their own version numbers (Project & Tooling Guide §2.2).
- **New code** uses the roci_ prefix, CONFIG_ROCI_* and the Roci F Prime namespace. Existing bac_ identifiers are renamed in a retrofit tracked as its own issue, not in passing.

4 Where This Applies

- bac-docs documentation covering the flight software stack.
- Code comments, namespaces and module names within bac-software.
- Codeberg issue and PR titles and descriptions referring to the stack as a whole.
- Discord and other community communication, in the project's plain-spoken voice (Identity Guide §2).

It does **not** apply to:

- bac-hardware or bac-structure, which are not part of the software stack.
- Third-party components – Zephyr, F Prime, libcsp, the ground segment software – which keep their upstream names.

5 Marks

Rocinante has its own marks at v2r1 – the horse head in a ring, the ROCINANTE wordmark over CUBESAT SOFTWARE STACK, and their stacked lockup – in bac-design `assets/` (`rocinante-*`), constructed as the Identity Guide §8 describes. They are BAC Marks under the BAC Trademark Policy and are not covered by the open licenses.

6 Version History

VERSION	DATE	CHANGE
0.2	2026-10-29	Moved from Google Drive into bac-design `guides/`. §1: the three node classes replace the flight and SBC segments; MicroMod and CM5 carriers; STM32F405 first, STM32U5 later; CSP over CAN. §2: CONFIG_ROCI_* and the Roci F Prime namespace. §3: settled by bac-software #105 – the repository stays bac-software, document IDs stay BAC-FSW-* and BAC-SBC-*, the bac_ retrofit as its own issue. §5: the Rocinante marks, replacing the open question on a logo. References to the current guide names. Title Case headings, American spelling.
0.1	2026-07-14	Initial note. Documented Rocinante/Roci naming rule, relationship to bac-software, and scope of application.

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